

MATH 4162/Math 6201

Numerical methods for differential equations

Fall 2025

Homework problem set #3

Show all necessary steps of working in the given space

Name: _____

ID: _____

Upload the assignment solution no later than

23:59 on Nov 17, 2025

Full marks 50

1. [25pts]

(a) Verify that $\{1, \cos x, \cos 2x, \dots\}$ is an orthogonal set on the interval $x \in [-\pi, \pi]$.

(b) Consider

$$\phi_n(x) = e^{2\pi i n x / L}$$

and verify that $\{\phi_n(x)\}$ is orthogonal on the interval $[0, L]$.

(c) The Leap-Frog scheme for

$$\frac{\partial u}{\partial t} + a(u) \frac{\partial u}{\partial x} = 0, \quad a > 0$$

using Fourier collocation method on a grid $\{x_j\}$ was given in the lecture notes. Determine the stability condition and show that

$$\Delta t \leq \frac{\Delta x}{\pi \max a(u)}.$$

(d) Consider a spectral differentiation of a PDE given as

$$\frac{\partial U}{\partial t} = \mathcal{D}U,$$

where $U = [u(x_0), \dots, u(x_{n-1})]^T$ denotes the solution of PDE as n -dimensional vector, and $\mathcal{D} \in \mathbb{R}^{n \times n}$ is the differentiation matrix. Let $E = (1/2)U^T U$ be the energy. Show that the energy is conserved if the differentiation matrix $\mathcal{D}$ is skew-symmetric.

(e) Consider the spectral differentiation matrix $\mathcal{D}$ given in lecture notes (lecture#13, page 4). If the 1D advection equation

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0, \quad a > 0$$

is discretized using the spectral matrix $\mathcal{D}$, show that the energy will not be conserved for any standard time integration method. Then propose a modification of $\mathcal{D}$ that ensures energy conservation?

2. [25pts]

(a) Consider the linear PDE:

$$\frac{\partial u}{\partial t} + 4 \frac{\partial u}{\partial x} = 0$$

Find the characteristic curves and draw three curves in the $x - t$ plane with starting points $(-5, 0)$, $(0, 0)$, and $(5, 0)$.

(b) Consider the nonlinear PDE:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0, \quad 0 < x < \pi$$

where $u(x, 0) = \sin(x_0)$ with x_0 being the starting location of the characteristic curves passing through (x, t) at $t = 0$.

Find the characteristic curves and draw three curves in the $x - t$ plane with starting points $(1/4, 0)$, $(1/2, 0)$, and $(3/4, 0)$.

Determine if any pair of the curves intersect. Find the point of intersection (x_c, t_c) and/or justify if there is no intersection.

(c) Consider the FV discretization of 1d advection equation. Substituting $Q_j^n = e^{ikj\Delta x}$ show that the amplification factor is

$$G = (1 - C) + Ce^{-ik\Delta x}$$

where $C = a\Delta t/\Delta x$.

(d) Consider the PDE

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = bu, \quad u(x, 0) = u_0(x)$$

Verify that the solution of this PDE is $u(x, t) = e^{bt}u_0(x - at)$. Consider the following scheme

$$Q_j^{n+1} = Q_j^n - \frac{a\Delta t}{\Delta x}(Q_j^n - Q_{j-1}^n) + \Delta t b Q_j^n.$$

Derive the leading order truncation error τ_j^n and show

$$\tau_j^n = \frac{1}{2}\Delta t^2(a^2u_{xx} - 2bau_x + b^2u) + \frac{1}{2}(a\Delta t\Delta x)u_{xx} + \text{higher order terms}$$

Show that the given scheme is Lax-Richtmyer stable provided $a\Delta t/\Delta x \leq 1$.

(e) [Math4162 only] Consider the hyperbolic system of linear PDES:

$$\frac{\partial p}{\partial t} + K_0 \frac{\partial u}{\partial x} = 0$$

$$\frac{\partial u}{\partial t} + \frac{1}{\rho_0} \frac{\partial p}{\partial x} = 0$$

where $K_0 = 4$ and $\rho_0 = 1$. The initial conditions are

$$(p, u)(x, 0) = \begin{cases} (p_l, u_l), & x < 0 \\ (p_r, u_r), & x > 0 \end{cases}$$

Compute the eigenvalues and (right and left) eigenvectors of the system matrix.

Write the jump $\Delta q = q_r - q_l$ as

$$\Delta q = \alpha_1 \phi_1 + \alpha_2 \phi_2$$

and find the values/expression of α_1 and α_2 .

(f) [*Math6201 only*] Consider the hyperbolic system of PDEs:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} (\rho u) = 0$$

$$\frac{\partial \rho u}{\partial t} + \frac{\partial}{\partial x} (\rho u^2 + p) = 0.$$

Find the eigenvalues of the system and determine whether the system is hyperbolic or not.

Show that

$$\frac{du}{dt} \pm \frac{a}{\rho} \frac{d\rho}{dt} = 0 \quad \text{along} \quad \frac{dx}{dt} = u \pm a,$$

where $a^2 = \frac{dp}{d\rho}$.